Slide 10: "Let's dive into how we actually built and tested the intelligence behind RequAI. We didn't want to just pick an algorithm blindly, so we designed and evaluated two completely different, complementary approaches. On one hand, we have Approach 1, which relies on Classical NLP. It's a very lightweight, highly interpretable pipeline that uses character TF-IDF features paired with linear classifiers. It's fast, transparent, and cheap to run. On the other hand, we wanted to see how far we could push the limits of language handling, so we built Approach 2. This is our Multilingual Deep Learning model. It adds language identification and uses transformer-based entity extraction so we can support all six targeted language formats. Let's look at how each one behaves under the hood."

Slide 11: "First, let's break down Approach 1. The traditional machine learning pipeline built on character TF-IDF features. For classifying user intent, we pass those features into a Linear Support Vector Machine, or SVM. Once we know what the user wants, job matching is handled mathematically using cosine similarity between the candidate's profile and the available job offers. If they ask a general question about the platform, the system pulls information directly from our curated ANEM knowledge base. We also baked in a fairness analysis step here by masking explicit identity signals right in the input text to prevent recommendations from pulling hidden biases. And performance-wise, it hit an 83% intent classification accuracy, and it successfully found answers in our knowledge base 100% of the time. The biggest perk here is that it requires absolutely zero GPU power to run, making it a low-cost fallback system."

Slide 12: "But as we noted earlier, Algerian job seekers don't just type in standard French or Modern Standard Arabic. They use Darija, they use Arabizi, and they constantly mix them together in a single sentence. That's where Approach 2 comes in—it's fully language-aware and executes in three sequential stages. Stage 1 is Language Detection. After testing several options, a Logistic Regression model gave us the best performance, hitting a 99.34% accuracy on our standard test set and maintaining a rock-solid 97.15% even on a difficult challenge set designed to trip it up. Once the language is locked in, we move to Stage 2: Intent Classification. We trained a model over a dataset of more than 45,000 examples spanning 18 distinct user intents, achieving a 99.74% accuracy across all six language variants. Finally, Stage 3 handles Entity Extraction.

We integrated XLM-RoBERTa to contextually pull out skills, job roles, and locations from highly unpredictable, cross-lingual text. This gives the assistant the true deep-learning backbone it needs to understand real, everyday Algerian dialect."

Slide 13: "Now, a great model is only useful if you can actually deploy it, so let's talk about our architecture and technical stack. We wanted RequAI to be fully containerized and easy to drop directly onto Wassit Online. For the frontend, we built a responsive, lightweight chat interface using React and Vite, designed to blend naturally right into the existing Wassit user experience. The backend is powered by Flask and orchestrated via Gunicorn. This acts as the brain that serves our machine learning models and coordinates the data flow. Speaking of data, we use SQLite to securely manage candidate profiles, job offers, and the core knowledge base text. When it comes to generating grounded, natural responses, we connect to Groq Cloud. This gives us access to high-speed hosted LLMs where we use strict prompt engineering to ensure the bot never hallucinates and only answers using our verified data. Finally, for production deployment, the whole ecosystem is containerized via Docker Compose and sits behind an Nginx reverse proxy to smoothly route traffic between the user and our services."

Slide 14: "Let's look at the hard numbers from our evaluation phase. The metrics speak for themselves: our deep-learning multilingual approach hit a 99.74% intent classification accuracy across those 18 learned intents. Our language detector proved its resilience by maintaining over 97% accuracy on our dialect-heavy challenge set. And our RAG pipeline achieved a perfect 100% retrieval rate for platform-related questions. Beyond the raw percentages, we validated two major qualitative goals. First, our cosine matching system effectively bypassed the rigid keyword limitations of the original platform, surfacing highly relevant jobs based on semantic meaning rather than exact word matches. Second, our bias analysis explicitly confirmed that when we mask identity variables like name or gender, the ranking algorithm provides a significantly fairer evaluation of candidate skills."

Slide 15: "To put it all in perspective, let's do a quick side-by-side comparison of our two approaches. As you can see on this chart, Approach 2—the Multilingual Deep Learning model—is the clear winner when it comes to raw performance. It beats Approach 1 in intent accuracy by over 16% and offers native, deep semantic support across all 6 language variations using XLM-RoBERTa instead of basic rule-based entity lookups. However, Approach 1 still has a very important place in our architecture. It costs almost nothing to run computationally, and its interpretability is incredibly high because it operates on transparent statistical weights rather than complex neural networks. Because of this, we chose Approach 2 as our primary production model, but we keep Approach 1 integrated as an incredibly fast, transparent fallback system if server demands spike."

Slide 16: "Of course, no research project is without its boundaries, and we want to be completely realistic about our current limitations. First, parts of our multilingual datasets were synthetically generated to cover gaps, which means they might not perfectly reflect every single quirk of real-world conversational text. Second, deep learning models naturally add higher compute costs and slight latency overhead compared to a basic search bar. Finally, our chatbot's knowledge is strictly bounded by the curated content currently loaded into our knowledge base. Looking ahead, our immediate next step is to launch real-world user studies directly on Wassit Online to see how actual job seekers interact with the bot in real time. We also plan to integrate voice input support and expand our linguistic models to capture even broader sub-dialects of Algerian Darija. Ultimately, we want to implement continuous learning loops so the system can organically improve itself based on live user feedback and successful hiring interactions."

Slide 17: "To wrap things up, let's look at the big picture. What we've built with RequAI is a practical, modern solution to a very real problem facing millions of Algerian job seekers and recruiters. By layering a conversational assistant onto Wassit Online, we've effectively bridged the gap between rigid, outdated keyword filtering and true semantic intelligence. We have successfully delivered an assistant that can comfortably understand six distinct linguistic inputs, extract profile criteria without human bias, match candidates based on actual meaning, and answer complex institutional questions reliably. Having this thoroughly validated across two distinct architectural approaches gives us immense confidence in the stability and readiness of this system."

Slide 18: With that, thank our supervisor, Dr. Telghamti Samira, and to all the members of the jury for your time. We are now completely open to your thoughts, feedback, and any questions you might have about RequAI. Thank you very much."